

Exp-31

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int pages[50], frames[10];
6     int n, f, i, j, k = 0;
7     int pageFaults = 0, pageHits = 0, found;
8
9     printf("Enter number of pages: ");
10    scanf("%d", &n);
11
12    printf("Enter the page reference string:\n");
13    for (i = 0; i < n; i++)
14        scanf("%d", &pages[i]);
15
16    printf("Enter number of frames: ");
17    scanf("%d", &f);
18
19    // Simulation
20    for (i = 0; i < n; i++)
21    {
22        found = 0;
23        for (j = 0; j < f; j++)
24        {
25            if (pages[i] == frames[j])
26            {
27                found = 1;
28                pageHits++;
29            }
30        }
31        if (found == 0)
32        {
33            // Page Fault
34            for (j = 0; j < f; j++)
35            {
36                if (frames[j] != -1)
37                {
38                    // Find the position to replace
39                    pos = j;
40                    min = frames[j];
41                }
42            }
43            // Replace the page
44            frames[pos] = pages[i];
45            pageFaults++;
46        }
47    }
48
49    printf("Total Page Faults = %d\n", pageFaults);
50    printf("Total Page Hits = %d\n", pageHits);
51}
```

input

Enter number of pages: 12
Enter the page reference string:
1 2 3 4 1 2 5 1 1 2 3 4 5
Enter number of frames: 5

Page	Frames	Status
1	1 - - -	Page Fault
2	1 2 - -	Page Fault
3	1 2 3 -	Page Fault
4	1 2 3 4 -	Page Fault
5	1 2 3 4 -	Page Hit
6	1 2 3 4 -	Page Hit
7	1 2 3 4 5	Page Fault
8	1 2 3 4 5	Page Hit
9	1 2 3 4 5	Page Hit
10	1 2 3 4 5	Page Hit
11	1 2 3 4 5	Page Hit
12	1 2 3 4 5	Page Hit

Total Page Faults = 5
Total Page Hits = 7

Exp-32

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int pages[50], frames[10];
6     int n, f, i, j, k;
7     int pageFaults = 0, pageHits = 0;
8     int found, pos, min;
9
10    printf("Enter number of pages: ");
11    scanf("%d", &n);
12
13    printf("Enter the page reference string:\n");
14    for (i = 0; i < n; i++)
15        scanf("%d", &pages[i]);
16
17    printf("Enter number of frames: ");
18    scanf("%d", &f);
19
20    // Simulation
21    for (i = 0; i < n; i++)
22    {
23        found = 0;
24        for (j = 0; j < f; j++)
25        {
26            if (pages[i] == frames[j])
27            {
28                found = 1;
29                pageHits++;
30            }
31        }
32        if (found == 0)
33        {
34            // Page Fault
35            for (j = 0; j < f; j++)
36            {
37                if (frames[j] != -1)
38                {
39                    // Find the position to replace
40                    pos = j;
41                    min = frames[j];
42                }
43            }
44            // Replace the page
45            frames[pos] = pages[i];
46            pageFaults++;
47        }
48    }
49
50    printf("Total Page Faults = %d\n", pageFaults);
51    printf("Total Page Hits = %d\n", pageHits);
52}
```

Enter number of pages: 12
Enter the page reference string:
1 2 3 4 1 2 5 1 1 2 3 4 5
Enter number of frames: 5

Page	Frames	Status
1	1 - - -	Page Fault
2	1 2 - -	Page Fault
3	1 2 3 -	Page Fault
4	1 2 3 4 -	Page Fault
5	1 2 3 4 -	Page Hit
6	1 2 3 4 -	Page Hit
7	1 2 3 4 5	Page Fault
8	1 2 3 4 5	Page Hit
9	1 2 3 4 5	Page Hit
10	1 2 3 4 5	Page Hit
11	1 2 3 4 5	Page Hit
12	1 2 3 4 5	Page Hit

Total Page Faults = 5
Total Page Hits = 7

Exp-33

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int pages[50], frames[10];
6     int n, f, i, j, k;
7     int pageFaults = 0, pageHits = 0;
8     int found, pos, farthest, next;
9
10    printf("Enter number of pages: ");
11    scanf("%d", &n);
12
13    printf("Enter the page reference string:\n");
14    for (i = 0; i < n; i++)
15        scanf("%d", &pages[i]);
16
17    printf("Enter number of frames: ");
```

Enter number of pages: 12
Enter the page reference string:
1 2 3 4 1 3 5 1 1 2 3 4 5
Enter number of frames:

Page	Frames	Status
1	1 - - -	Page Fault
2	1 2 - -	Page Fault
3	1 2 3 -	Page Fault
4	1 2 3 4 -	Page Fault
1	1 2 3 4 -	Page Hit
3	1 2 3 4 -	Page Hit
5	1 2 3 4 5	Page Fault
1	1 2 3 4 5	Page Hit
1	1 2 3 4 5	Page Hit
2	1 2 3 4 5	Page Hit
3	1 2 3 4 5	Page Hit
4	1 2 3 4 5	Page Hit

Total Page Faults = 5
Total Page Hits = 7

Exp-34

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int records[50];
6     int n, i, search, found = 0;
7
8     printf("Enter the number of records: ");
9     scanf("%d", &n);
10
11    printf("Enter the records:\n");
12    for (i = 0; i < n; i++)
13    {
14        records[i] = i + 1;
15        printf("Record %d: ", records[i]);
16    }
17
```

Enter the number of records: 5
Enter the records:
Record 1: Record 2: Record 3: Record 4: Record 5:
Enter the record number to access: 104

Reading records sequentially...

Reading Record 1
Reading Record 2
Reading Record 3
Reading Record 4
Reading Record 5
Record 104 not found.

Exp-35

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int indexBlock[50];
6     int n, i, block, found = 0;
7
8     printf("Enter the number of file blocks: ");
9     scanf("%d", &n);
10
11     printf("Enter the block numbers:\n");
12     for (i = 0; i < n; i++)
13     {
14         scanf("%d", &indexBlock[i]);
15     }
16
17     printf("\nIndex Block\n");
18     printf("-----\n");
19
20     for (i = 0; i < n; i++)
21     {
22         printf("Index %d -> Block %d\n", i, indexBlock[i]);
23     }
24
25     printf("\nEnter the block number to access: ");
26     scanf("%d", &block);
27
28     found = 0;
29     for (i = 0; i < n; i++)
30     {
31         if (indexBlock[i] == block)
32         {
33             found = 1;
34             break;
35         }
36     }
37
38     if (found == 1)
39     {
40         printf("Block %d is present in the file.\n", block);
41     }
42     else
43     {
44         printf("Block %d is not present in the file.\n", block);
45     }
46 }
```

input

Enter the number of file blocks: 5
Enter the block numbers:
12 10 3 35 2 90

Index Block

Index 0 -> Block 12
Index 1 -> Block 10
Index 2 -> Block 3
Index 3 -> Block 35
Index 4 -> Block 2

Enter the block number to access: Block 90 is not present in the file.

Exp-36

```
39 end = blocks[n - 1].blockNo;
40
41 printf("\n--- Linked Allocation ---\n");
42 printf("Directory Information:\n");
43 printf("First Block : %d\n", start);
44 printf("Last Block : %d\n", end);
45
46 printf("\nFile Block Chain:\n");
47
48 current = start;
49
50 while (current != -1) {
51     printf("%d", current);
52     current = blocks[current].next;
53 }
```

input

Enter number of blocks in the file: 4

Enter the block numbers and next block pointers:

Block 1 number: 21
Next block number: 23

Block 2 number: 44
Next block number: 32

Block 3 number: 67
Next block number: 43

Block 4 number: 22
Next block number: -1 (End of file)

--- Linked Allocation ---
Directory Information:
First Block : 21
Last Block : 22

File Block Chain:
21 -> 23 -> Invalid block

Exp-37

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int main()
5 {
6     int n, i, head, total_movement = 0;
7     int request[100];
8
9     printf("Enter the number of disk requests: ");
10    scanf("%d", &n);
11
12    printf("Enter the disk request queue:\n");
13    for (i = 0; i < n; i++)
14    {
15        scanf("%d", &request[i]);
16    }
17
18    printf("Enter the initial head position: ");
19    scanf("%d", &head);
20
21    printf("\nFCFS Disk Scheduling\n");
22
23    printf("Enter the number of disk requests: 3\n");
24    printf("Enter the disk request queue:\n");
25    98
26    83
27    5
28    printf("Enter the initial head position: 1\n");
29
30    printf("FCFS Disk Scheduling\n");
31    printf("Disk head movement:\n");
32    printf("1 -> 98 -> 183\n");
33
34    printf("Total Head Movement = 320 cylinders\n");
```

Exp-38

```
90     total_movement += head;
91     head = 0;
92     printf(" -> %d", head);
93 }
94
95 /* Reverse direction */
96 for (i = 0; i < n; i++)
97 {
98     if (request[i] > head)
99     {
100         total_movement += abs(request[i] - head);
101         head = request[i];
102         printf(" -> %d", head);
103     }
104 }
105 }
106
107 printf("\nTotal Head Movement = %d cylinders\n", total_movement);
108
109 return 0;
110
```

input

```
Enter number of disk requests: 5
Enter disk request queue:
12
34
67
89
90
Enter initial head position: 3
Enter disk size: 23
Enter direction (1 for high, 0 for low): 43

SCAN Disk Scheduling
Head movement: 3 -> 0 -> 12 -> 34 -> 67 -> 89 -> 90
Total Head Movement = 93 cylinders
```

Exp-39

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int main()
5 {
6     int request[100], n, head, disk_size;
7     int i, j, temp, total_movement = 0;
8
9     printf("Enter number of disk requests: ");
10    scanf("%d", &n);
11
12    printf("Enter disk request queue:\n");
13    for (i = 0; i < n; i++)
14        scanf("%d", &request[i]);
15
16    printf("Enter initial head position: ");
17    scanf("%d", &head);
18
19    printf("Enter disk size: ");
20    scanf("%d", &disk_size);
21
22    /* Sort requests */
23
24    Enter number of disk requests: 3
25    Enter disk request queue:
26    12
27    45
28    89
29
30    Enter initial head position: 4
31    Enter disk size: 5
32
33    C-SCAN Disk Scheduling
34    Head movement: 4 -> 12 -> 45 -> 89 -> 4 -> 0
35    Total Head Movement = 174 cylinders
```

Exp-40

```
1 #include <stdio.h>
2
3 int main()
4 {
5     printf("Linux File Access Permissions\n");
6
7     Linux File Access Permissions
8     -----
9
10    Types of Users:
11    1. Owner (u) - User who owns the file
12    2. Group (g) - Users belonging to the group
13    3. Others (o) - All other users
14
15    File Access Permissions:
16    r - Read (4)
17    w - Write (2)
18    x - Execute (1)
19
20    Example Permission: rwxr-xr--
21    Owner : rwx - Read, Write, Execute
22    Group : r-x - Read, Execute
23    Others : r-- - Read only
24
25    Numeric Representation:
26    rwx = 7
27    r-x = 5
28    r-- = 4
29
30    Example Command:
31    chmod 754 file.txt
32
33    Meaning of chmod 754:
34    Owner : 7 -> rwx
35    Group : 5 -> r-x
36    Others : 4 -> r--
```

